

DATA VORTEX A'26

ROUND 3 • ANALYSIS NOTEBOOK

Public Reaction to a Trending Social Issue
Focus: Cybercrime & Digital Exclusion

Notebook-style analytical report with verified summary statistics, charts, methodology notes, and reproducibility appendix.

Prepared from the analysis details shared in this conversation and the uploaded collection notebook.
See data-source and validation notes below.

1. Project objectives

This analysis examines news coverage related to Cybercrime and Digital Exclusion as issue-related information activity. It summarizes collected article records, predicted sentiment and topic labels, and publication-date patterns. It does not directly measure public opinion or social-media engagement.

2. Dataset overview & collection

Metric	Reported value
Combined unique articles	234
Fields	13 columns
Duplicate URLs after cleaning	0
Missing values	0
Focus groups	Cybercrime; Digital Exclusion; Other
Collection source	Google News RSS

The shared project context describes two collection batches merged by URL, retaining batch provenance. The uploaded notebook contains code for collection, comparison, timestamp checks, summaries, and charts. This PDF does not claim a fresh execution of that notebook.

3. Data cleaning & date validation

The reported cleaned dataset has 234 unique URLs. Two records have a suspicious published_at value of 1970-01-01 and should remain in the master dataset while being excluded from temporal analysis until independently verified. These records were identified as an Interpol cybercrime article and a Jamaica Information Service legislation article.

Date field	Reported range / handling
published_at	1970-01-01 to 2026-09-21 07:45:38 UTC; 1970 values flagged
collected_at	2026-09-21 02:59:38 to 07:48:21 UTC
Temporal subset	232 records after excluding the two epoch-date records

Keep publication time and collection time distinct. A publication-date trend describes when indexed articles were published—not when the public reacted.

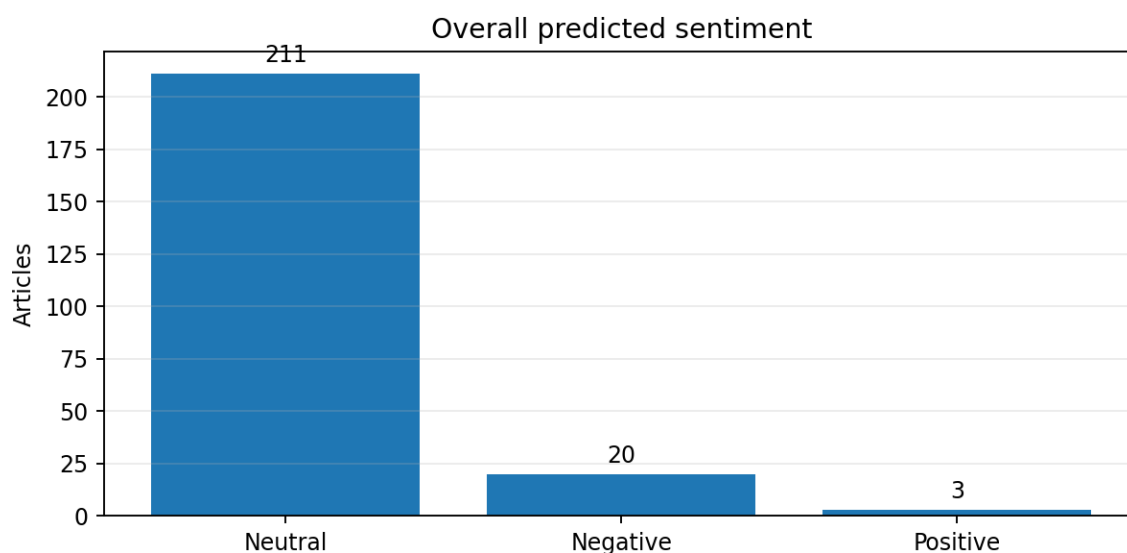
4. Exploratory data analysis

Focus group	Articles	Share
Digital Exclusion	119	50.9%
Cybercrime	97	41.5%
Other / neither	18	7.7%

The rounded shares sum to approximately 100.1% due to rounding.

5. Predicted sentiment

Predicted label	Count	Share
Neutral	211	90.2%
Negative	20	8.5%
Positive	3	1.3%



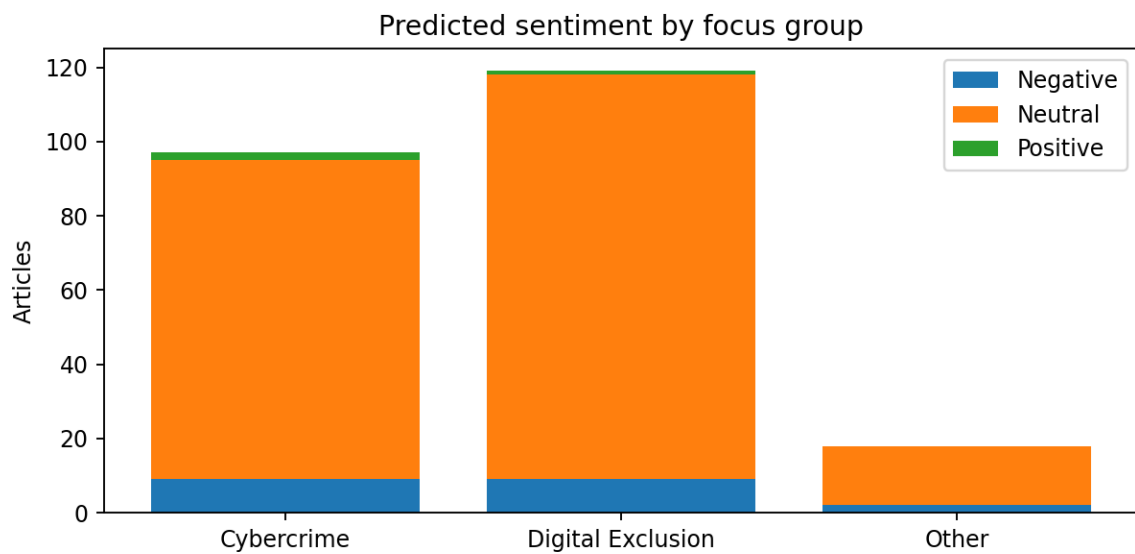
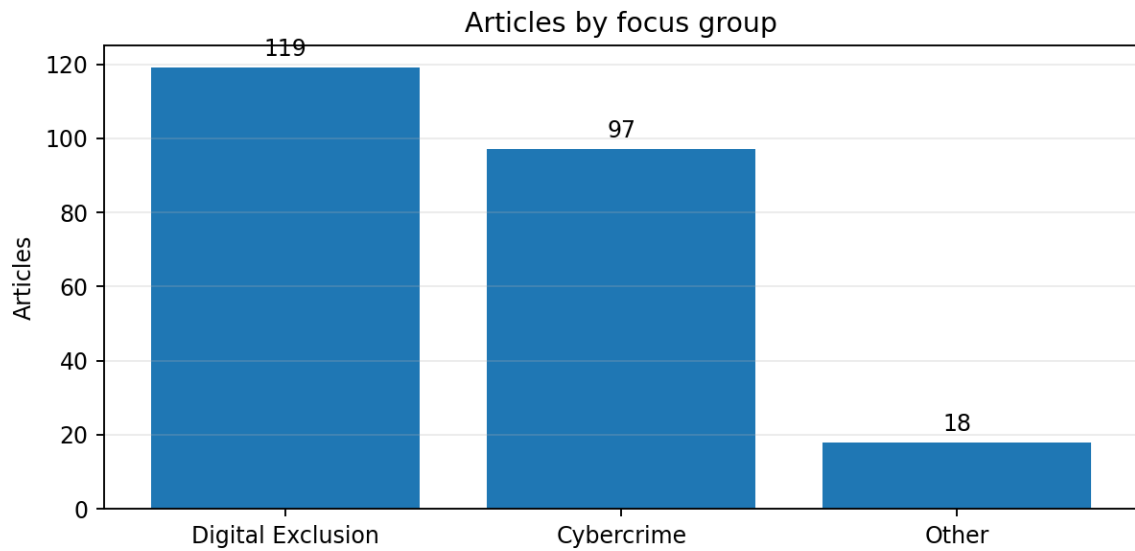
Interpretation: predictions are strongly concentrated in Neutral. This is a model output over article text, not a direct measure of public sentiment. The supplied summary does not establish held-out accuracy or human-label agreement.

6. Topic classification

Predicted topic	Count
Community_Discussion	230
Technical_Issues	2
Account_Security	1
Feature_Feedback	1

Extreme class imbalance makes the topic distribution difficult to interpret. Treat these as predicted labels; manually validate examples and report class-wise performance before drawing topic conclusions.

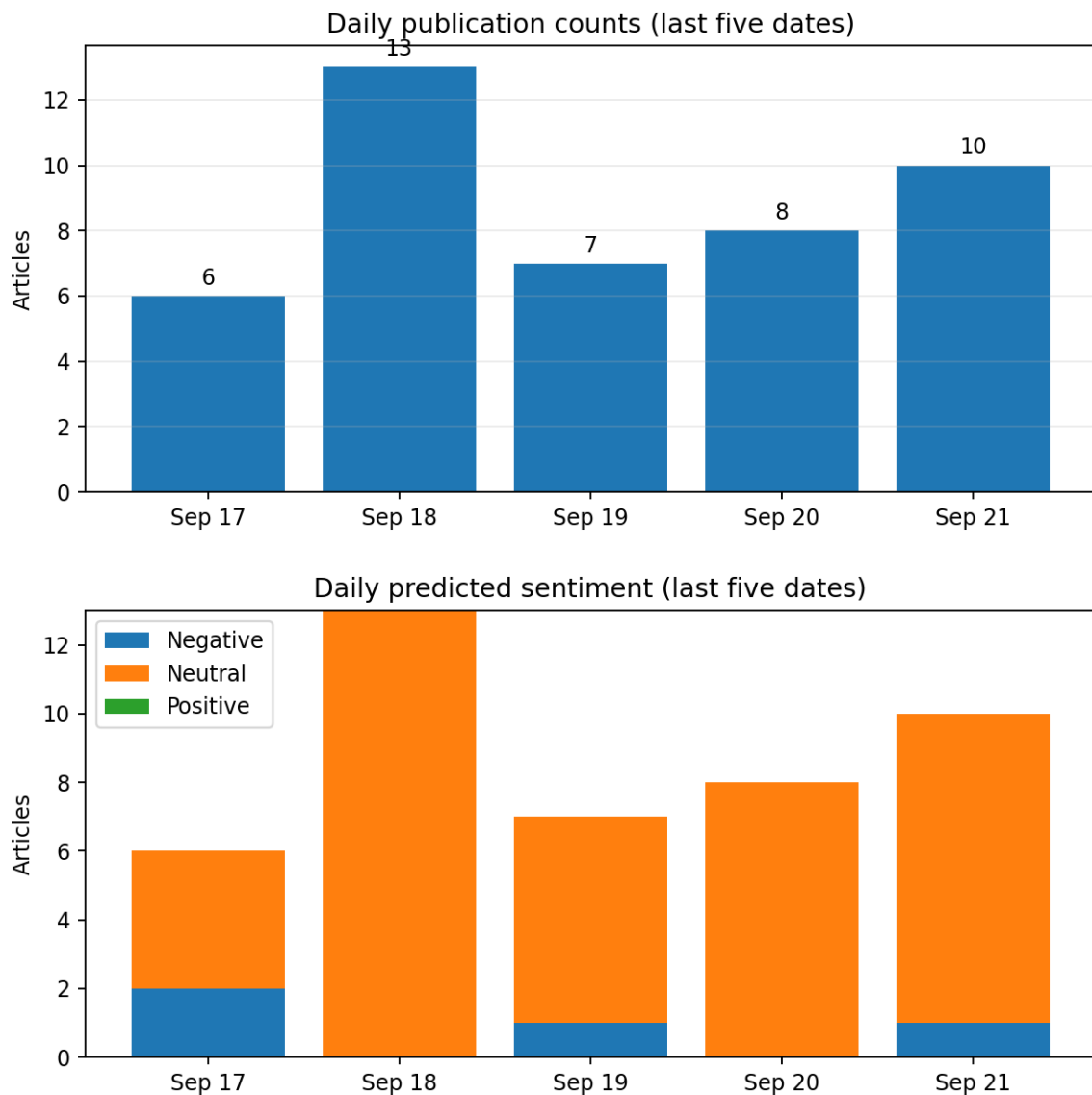
7. Cybercrime vs Digital Exclusion



Within-group predicted sentiment: Cybercrime—Negative 9.3%, Neutral 88.7%, Positive 2.1%; Digital Exclusion—Negative 7.6%, Neutral 91.6%, Positive 0.8%. Other—Negative 11.1%, Neutral 88.9%, Positive 0%. Percentages may differ slightly from 100 due to rounding.

8. Publication activity

There are 121 distinct publication dates in the reported date analysis. The final five dates have the following article counts:



Daily sentiment counts: Sep 17: 2 Negative, 4 Neutral; Sep 18: 13 Neutral; Sep 19: 1 Negative, 6 Neutral; Sep 20: 8 Neutral; Sep 21: 1 Negative, 9 Neutral. Positive predictions were zero on these dates.

A daily article-count increase is not equivalent to a social engagement spike. Two snapshots cannot establish continuous monitoring, sentiment change, or causality.

9. Key findings

- The dataset is split mainly between Digital Exclusion (119) and Cybercrime (97), with 18 records outside those focus groups.
- Predicted sentiment is overwhelmingly Neutral (211/234).
- Topic predictions are highly imbalanced: 230 of 234 are Community_Discussion.
- Two epoch-like publication dates require verification and exclusion from temporal charts pending validation.
- The current RSS dataset measures news coverage, not likes, comments, shares, views, or survey responses.

10. Limitations & ethical considerations

RSS coverage is shaped by search terms, publisher indexing, language, geography, and source availability. Model predictions can misclassify nuance, sarcasm, headlines, and context. Avoid inferring population-wide opinion, causal triggers, or individual beliefs. If the competition requires direct public reaction, add ethically

collected, permitted engagement or survey data and document sampling limitations.

11. Recommendations & conclusion

Manually validate a stratified sample of sentiment and topic predictions; evaluate models on a held-out labeled set; document search queries and snapshot times; preserve raw records; maintain date-quality flags; and align every report statistic with the final CSV. Reframe conclusions as patterns in indexed news coverage unless direct reaction data are added.

Conclusion: The collected news sample shows substantial coverage of both focus areas, predominantly Neutral model predictions, and an imbalanced topic classifier. These results support descriptive news-coverage analysis, but do not independently establish public reaction.

12. Appendix — illustrative reproducible analysis code

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("DATA_VORTEX_R3_combined_clean.csv")
assert "url" in df.columns
assert df["url"].nunique() == len(df), "Duplicate URLs detected"

df["published_at"] = pd.to_datetime(
    df["published_at"], errors="coerce", utc=True
)
df["collected_at"] = pd.to_datetime(
    df["collected_at"], errors="coerce", utc=True
)

# Preserve all records; exclude suspicious dates only for temporal analysis
date_ok = (
    df["published_at"].notna()
    & (df["published_at"] >= pd.Timestamp("2000-01-01", tz="UTC"))
)
time_df = df.loc[date_ok].copy()

print("Rows:", len(df))
print("Unique URLs:", df["url"].nunique())
print("Missing values:", int(df.isna().sum().sum()))
print(df["sentiment_prediction"].value_counts(dropna=False))
print(df["topic_prediction"].value_counts(dropna=False))

# Publication activity (not social engagement)
time_df["publication_date"] = time_df["published_at"].dt.date
daily_counts = time_df.groupby("publication_date").size()
daily_counts.plot(kind="line", marker="o")
plt.title("Daily indexed news articles")
plt.xlabel("Publication date (UTC)")
plt.ylabel("Article count")
plt.tight_layout()
plt.show()
```

Appendix note: this is a concise reproducibility example, not a replacement for the complete original collection/model pipeline. The notebook was inspected structurally; no claim is made that all cells were executed in a fresh runtime.